LESSON PLAN

LSF, STABILITY, A/C PERFORMANCE (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	27
Total No. of Lesson Plans	33	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Lateral stability: forces and Moments, effect of wing dihedral, effects of slide slip, induced roll and induced yaw.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Lateral stability: forces and Moments, effect of wing dihedral, effects of slide slip, induced roll and induced yaw. Factors effecting lateral stability. Wing sweepback effect. Dihedral effect
5. Specific Objectives	6.1.4.1 Trainee will be able to describe aircraft stability in all the three axis.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of Lateral Stability concepts using PPT diagrams; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the three axes of an aircraft?
(b) Define lateral stability in your own words.
(c) Which axis does lateral stability primarily control?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Lateral Stability • Forces and Moments involved • Effect of Wing Dihedral • Wing Sweepback Effect	L & Ds	PPT, IB	15 min
(b)	• Effects of Sideslip • Induced Roll Mechanism • Induced Yaw Mechanism	L & Ds	PPT, PDF	20 min
(c)	• Factors Affecting Lateral Stability • Summary of Stability Characteristics • Transition to Dynamic Stability	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how wing dihedral contributes to lateral stability.
(b) Describe the relationship between sideslip and induced roll.
(c) How does wing sweepback affect lateral stability?

12. Home Assignment and Follow Up	Prepare a brief note comparing the effects of dihedral and sweepback on lateral stability, referencing concepts from LP 28.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Longitudinal dynamic stability, lateral dynamic stability.	Refer to the master syllabus reference list	LP 28

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min